Strategic Architecture and Implementation Blueprint for an Autonomous LLM-Based Material Selection System

Executive Summary

The intersection of advanced artificial intelligence and mechanical engineering presents transformative opportunities for localized manufacturing and rapid prototyping. In developing economies with dispersed rural populations, such as Kenya, the ability to locally manufacture and maintain critical infrastructure—specifically decentralized renewable energy systems—is hampered by complex material science constraints and supply chain bottlenecks[1]. To democratize engineering design and facilitate local manufacturing, there is a pressing need for intelligent, autonomous systems capable of translating non-technical requirements into precise material specifications.

This comprehensive report outlines the architecture, theoretical framework, and execution roadmap for an advanced, dual-mode Large Language Model (LLM) powered material selection tool. Designed to fulfill the rigorous requirements of a final-year mechanical engineering capstone project over a rapid four-month timeline, this system utilizes Retrieval-Augmented Generation (RAG) to eliminate AI hallucinations and provide mathematically rigorous material recommendations. To synthesize the project's foundational ideas, the report integrates a highly detailed case study: the design, material selection, and surface optimization for 3D-printed micro-hydro turbine blades tailored for Kenyan rural electrification. By combining the development of an autonomous, offline-capable AI tool with a high-impact mechanical engineering application, this blueprint provides an exhaustively detailed framework that bridges software engineering, materials science, and fluid dynamics.

The Kenyan Renewable Energy Landscape and Decentralized Manufacturing

To contextualize the necessity of localized manufacturing and the proposed AI material selection tool, it is imperative to analyze the macroeconomic and infrastructural realities of the target deployment environment. Kenya's electricity generation relies heavily on large-scale hydro, which accounts for nearly half of the grid's capacity but remains highly vulnerable to climate change and drought conditions[3]. Furthermore, thermal fossil fuel generation still constitutes a significant portion of the energy mix, exacerbating import costs and environmental degradation[3].

Despite national grid expansion efforts, rural electrification rates remain critically low, with the vast majority of un-electrified rural households relying on kerosene for lighting and biomass for thermal energy[3]. The national grid extension often proves economically unviable due to the

dispersed nature of rural populations and characteristically low power consumption metrics in these regions[6].

The Micro-Hydro Imperative

In response to the limitations of grid extension and diesel generation, decentralized renewable energy solutions have emerged as the most viable alternative. Micro-hydrokinetic river turbine technology (systems generating up to 100 kW without damming major waterways) offers a decentralized, continuous renewable energy source that bypasses the exorbitant costs of national grid extension[4]. The potential for small, mini, and micro-hydro systems in Kenya is estimated at approximately 3,000 MW nationwide, spread across major drainage basins such as the Tana River and Lake Victoria catchments[4].

Case studies, such as the Tungu-Kabiri pico-hydro project in the Mount Kenya region, demonstrate that community-managed micro-hydro systems are highly effective for rural electrification, providing mechanical power for agro-processing and electrical power for households and vocational centers[8]. However, the traditional manufacturing of turbine blades relies on expensive molds, tooling, and imported components[10]. Maintenance in remote areas is severely delayed by supply chain friction, where a simple blade fracture can render a community without power for months while awaiting imported replacement parts[7].

The Additive Manufacturing Disruption

Additive manufacturing (3D printing) disrupts this centralized, import-reliant model by enabling the creation of complex geometries layer by layer from digital files[2]. By transitioning to 3D printing, mold manufacturing costs can be reduced by up to 50%, and the need for expansive centralized infrastructure is bypassed[12]. The global market for 3D-printed turbine components is rapidly expanding, indicating the industrial viability of the technology[12].

Crucially, the emergence of indigenous 3D printing technology in Kenya establishes a foundation for decentralized hardware manufacturing[13]. Local innovators have successfully constructed multi-head 3D printers utilizing locally sourced components, proving that the hardware necessary for on-site fabrication is accessible[13]. If a structural issue or breakage occurs at a remote micro-hydro site, an engineer equipped with a portable 3D printer and the appropriate polymer filament can diagnose, print, and replace the turbine blade on-site within hours.

The primary bottleneck is no longer the hardware; it is the highly specialized knowledge required to select the correct materials and printing parameters to ensure the replacement part survives the rigorous mechanical and hydrodynamic loads of a river environment. This is the precise knowledge gap the proposed AI material selection tool is designed to bridge.

Theoretical Foundations of Mechanical Material Selection

At the inception of any mechanical design process, material selection plays a paramount role, dictating the component's performance, cost, durability, aesthetics, and manufacturability[14]. The intrinsic properties of a material determine its response to physical stresses, thermal loads, and environmental degradation over its life cycle[14]. Selecting the wrong material for a

3D-printed turbine blade could lead to catastrophic fatigue failure, excessive water absorption, or catastrophic delamination under load.

The Ashby Optimization Methodology

The universally recognized standard for rigorous material selection in mechanical design is the Ashby method. Developed by Michael F. Ashby, this framework optimizes material choice based on mathematical performance indices rather than isolated, singular properties[16]. A performance index is a quantitative combination of material properties that governs the overall performance of a component under specific functional constraints[16].

The derivation of an Ashby performance index begins by explicitly defining the function, objective, and constraints of the mechanical component[16]. As a foundational example, consider the design of a lightweight, stiff beam, a geometry analogous to the internal spars of a turbine blade. The stiffness S of a simply supported beam with a central load F, central deflection δ, length L, Young's modulus E, and second moment of area I is defined by classical mechanics:

$$S = \frac{F}{\delta} = \frac{C_1 E I}{L^3}$$

where C_1 is a constant depending on the load distribution ($C_1 = 48$ for 3-point bending)[16]. For a beam of square cross-section with area A, the second moment of area is $I = A^2/12$. The mass m of the beam is defined by its density ρ, cross-sectional area A, and length L:

$$m = AL\rho$$

To minimize mass while maintaining a required baseline stiffness, the free variable (the area A) is eliminated from the equations, yielding a mass equation dependent purely on the geometric constraints and the material properties:

$$m = \left(\frac{12SL^3}{C_1}\right)^{1/2} L \left(\frac{\rho}{E^{1/2}}\right)$$

To achieve the engineering objective of minimizing the component's mass, the material-dependent term in the equation must be minimized. Conversely, the performance

index M, which is to be maximized by the designer, is the inverse of this term:

$$M = \frac{E^{1/2}}{\rho}$$

Traditionally, mechanical engineers utilize Ashby charts—log-log plots of properties such as Young's modulus versus density—to draw selection lines of constant M (e.g., lines with a slope of 2 on a logarithmic scale) to visually identify optimal material families that satisfy the equation[16].

This methodology extends far beyond simple beams. For instance, if the engineering task involves a flywheel for energy storage (relevant to wind turbine generator inertia), the objective is to maximize kinetic energy per unit mass. The kinetic energy KE is limited by the maximum radial stress before the material yields (σ_y). The resulting performance index to maximize is:

$$M = \frac{\sigma_y}{\rho}$$

Similarly, designing for thermal diffusion (e.g., heat dissipation in the generator housing) requires maximizing thermal diffusivity, leading to indices incorporating thermal conductivity (λ), density (ρ), and specific heat capacity (c_p)[21].

The Barrier of Traditional Selection Methods

While mathematically sound and exhaustive, the traditional Ashby method presents substantial barriers for non-technical operators, field technicians, or localized manufacturers lacking formal metallurgical or polymeric training. The utilization of Ashby charts requires a deep understanding of complex boundary constraints, geometric shape factors (which measure the bending efficiency of a cross-sectional shape), and multi-objective calculus[14].

Furthermore, when designing for fused deposition modeling (FDM) additive manufacturing, the anisotropic nature of the printed materials renders standard two-dimensional Ashby plots insufficient[22]. A 3D-printed part exhibits significantly different tensile strengths depending on whether the load is applied parallel or perpendicular to the printed layer lines[23]. Additionally, environmental constraints specific to the application—such as UV degradation in solar applications or water saturation in micro-hydro applications—are not easily plotted on standard mechanical Ashby charts[22].

Therefore, to democratize local manufacturing, an intelligent, conversational interface is required to bridge the gap between human intent and rigorous engineering mathematics. The system must act as an automated materials engineer, capable of deducing the correct Ashby

indices from plain-text descriptions of the required work.

Architectural Design of the AI Material Selection Tool

To fulfill the capstone project's requirement of creating an AI material selection tool capable of interacting with non-technical users, a highly specific software architecture must be established. The user must be able to input a prompt such as, "I want to make an impeller that will carry a heavy load in a river, but I do not know the exact weight." The AI must then translate this layman's request into engineering parameters, calculate the optimal materials, output the material specifications, and provide structural integrity constraints in a fully contained environment.

The Dual-Mode Architecture: Standalone LLM vs. Cloud API

The foundational architectural decision lies in choosing between a cloud-dependent Application Programming Interface (API), such as OpenAI's GPT-4, and a standalone, locally hosted Large Language Model (LLM). The project requirements stipulate a preference for a standalone system but request a decision on whether it should be strictly AI-powered (cloud) or local.

The optimal engineering solution is a **Dual-Mode Architecture** with a local-first default. For deployment in regions like rural Kenya, where engineers frequently travel to remote micro-hydro or wind sites for maintenance and repair, reliance on a continuous, high-speed internet connection is a fatal flaw[3]. If the tool requires a cloud API, it becomes useless exactly when and where it is needed most: during on-site field repairs. Therefore, the standalone local LLM is the mandatory primary operational mode.

Recent advancements in model quantization allow highly capable LLMs, such as Llama-3 8B, Mistral, or Qwen 32B, to run entirely on consumer-grade hardware without requiring server-farm computational power[24]. The hardware requirements for a standalone LLM scale linearly with the parameter count and the quantization level. An 8-billion parameter model utilizing 4-bit quantization (Q4_K_M) requires approximately 4 to 8 GB of Video RAM (VRAM), making it easily executable on a standard engineering laptop equipped with a mid-range GPU or even a modern Apple Silicon MacBook utilizing Metal framework acceleration[24].

Beyond offline capability, a local LLM guarantees total proprietary data privacy, incurs zero recurring API token costs, and allows for deep system customization. Open-source ecosystems, such as Ollama and llama.cpp, provide drop-in OpenAI-compatible API servers[25]. However, to provide maximum flexibility, the architecture will be designed as an API-agnostic platform. By utilizing standard environment variables, the user can toggle the system to connect to a cloud API (like OpenAI) when high-speed internet is available in an urban office setting, leveraging the massive parameter counts of GPT-4 for highly abstract conceptual design, and then switch seamlessly back to the local llama.cpp instance for offline field deployment.

Overcoming LLM Deficiencies via Retrieval-Augmented Generation (RAG)

Standard LLMs, whether cloud-based or local, are fundamentally probabilistic text generators designed to predict the next most likely token in a sequence[30]. They lack intrinsic structured

mathematical reasoning and are highly prone to "hallucinations"—generating text that sounds plausible but is factually incorrect[31]. In engineering design, an AI hallucinating a material's yield strength, failing to account for anisotropic layer adhesion, or suggesting an incompatible manufacturing process can lead to catastrophic physical failure and severe safety hazards[15]. Furthermore, foundational LLMs cannot natively ingest dynamic, up-to-date information, such as the introduction of a novel 3D-printing filament formulation released after the model's training cut-off date[32].

While techniques like Supervised Fine-Tuning (SFT), Continued Pretraining (CPT), Direct Preference Optimization (DPO), and Odds Ratio Preference Optimization (ORPO) can align a model's tone and behavior, they do not inherently solve the hallucination of hard numerical data[33].

To completely mitigate this risk, the standalone LLM must not act as a primary knowledge source. Instead, it must serve solely as a natural language reasoning and translation engine within a Retrieval-Augmented Generation (RAG) framework[31]. The RAG architecture anchors the LLM's outputs to a verified, explicitly defined database of material properties, Ashby charts, and engineering principles, ensuring factual accuracy[32].

Building the Vector Database with ChromaDB

The foundation of the RAG system is the vector database, which stores the engineering knowledge. The proposed architecture utilizes ChromaDB, a highly optimized, open-source vector database tailored specifically for AI memory systems and Python integration[37].

The database construction begins by curating a structured and unstructured knowledge base. This includes scraping and formatting material data sheets (MDS) for all available 3D printing filaments, digitizing chapters of Ashby material selection textbooks, inputting additive manufacturing constraints, and compiling environmental data regarding UV exposure and water saturation[14].

This textual data is passed through an embedding model (such as a lightweight local model like all-MiniLM-L6-v2 via HuggingFace or sentence-transformers)[28]. The embedding model converts the text into high-dimensional numerical vectors. These embeddings represent the semantic "essence" or latent-space position of the data[37]. For instance, texts discussing "high tensile strength," "water resistance," and "PETG-CF" will cluster closely together in the multidimensional space[37]. ChromaDB stores these embeddings persistently on the local disk, allowing for rapid, offline semantic search[28].

Orchestration via LangChain

LangChain serves as the neurological orchestration framework of the tool, managing the complex flow of data between the user's non-technical prompt, the ChromaDB vector database, and the local LLM[28]. LangChain allows for the creation of intricate prompt templates and reasoning chains.

The NLP Translation Layer is the system's most critical software feature. When a non-technical user inputs a prompt—such as, "I want to make a blade that will push water, carry a heavy load, and sit in a river all day, but I don't know the exact weight"—the following sequence occurs:

1. **Intent Parsing:** LangChain instructs the LLM to parse the layman's terms. "Carry a heavy

load" is translated to a requirement for high yield strength (σ_y) and fracture toughness. "Sit in a river all day" is translated to high hydrolysis resistance, low water absorption rates, and corrosion resistance. "Blade that will push water" suggests fatigue resistance due to cyclical hydrodynamic loading.

2. **Query Embedding:** LangChain converts these translated engineering requirements into a vector using the identical embedding model used to create the database[35].
3. **Semantic Search:** LangChain queries ChromaDB using a cosine similarity search to retrieve the top K most mathematically and contextually relevant text chunks from the curated engineering database[37].
4. **Prompt Augmentation:** The retrieved factual data (e.g., exact yield strengths in MPa, density, water absorption percentages of various specific plastics) is injected into a strict system prompt template.
5. **Generation and Guardrails:** The standalone LLM receives the augmented prompt. Bound by strict system instructions to rely *only* on the provided ChromaDB context, the LLM generates a comprehensive, accurate response[31]. If the context does not contain the answer, the LLM is programmed to explicitly state the knowledge gap rather than hallucinate a response.

User Interface Integration with Streamlit

To fulfill the requirement that the tool must easily interact with non-technical users, a graphical user interface (GUI) is developed using Streamlit[39]. Streamlit is an open-source Python library that facilitates the rapid deployment of data science web applications with straightforward syntax[39].

Using the streamlit-chromadb-connection package, the frontend seamlessly interfaces with the ChromaDB backend, providing a clean, conversational chat interface[28]. The Streamlit interface will feature text input fields for the user's prompt, toggle switches to select between the local LLM and the cloud API (the dual-mode function), and visual displays of the output. The output will be formatted using Markdown to present the material specifications, structural integrity parameters, and manufacturing constraints in concise, easy-to-read data tables rather than dense blocks of text.

Case Study Application: Designing the 3D Printed Micro-Hydro Turbine Blade

To rigorously validate the effectiveness of the RAG-enabled LLM tool, the system must be tested against a highly complex, real-world engineering challenge. The localized additive manufacturing of micro-hydro turbine blades in Kenya provides an optimal test vector, as it encompasses complex mechanical loads, severe environmental conditions, and specific manufacturing constraints.

In this validation scenario, the user prompts the newly developed AI tool with the parameters for a micro-hydro turbine blade intended for deployment in a rural Kenyan river. The AI must navigate multiple conflicting constraints: high tensile strength, high fatigue resistance, minimal

water saturation degradation, printability via local Fused Deposition Modeling (FDM) systems, and hydrodynamic surface efficiency.

Material Analysis and Selection

The RAG system queries the ChromaDB database, retrieving comparative data on standard 3D printing FDM materials, specifically Polylactic Acid (PLA), Polyethylene Terephthalate Glycol (PETG), and their Carbon Fiber (CF) reinforced variants.

The AI tool analyzes the retrieved structural parameters, summarized in the table below, which represents the structured output the Streamlit GUI would present to the user:

Material Type	Tensile Strength	Environmental Resistance	Fatigue & Impact Dynamics	Additive Manufacturing Constraints
PLA	50 – 60 MPa	Poor. Rapid degradation in water and UV exposure.	High stiffness, but low fatigue limit and brittle.	Easiest to print, low warping, excellent layer adhesion.
PETG	40 – 50 MPa	Good chemical and water resistance. Moderate UV.	High impact resistance, ductile under load.	Prone to stringing, requires higher hot-end temperatures.
PETG-CF	45 – 55 MPa	Excellent water resistance, high UV resistance.	High stiffness, increased fatigue resistance, low creep.	Requires hardened steel nozzle due to abrasive carbon fibers.

While pure PLA exhibits a higher base tensile strength (up to 60 MPa) than pure PETG (up to 50 MPa), the system's NLP logic algorithms correctly prioritize the environmental constraints translated from the user's prompt[43]. The retrieved literature dictates that 3D-printed PLA specimens suffer a significant, catastrophic degradation of their tensile strength and modulus (by up to 28.4% and 7.2%, respectively) when placed in a water-saturated condition[22]. This renders PLA entirely unsuitable for submerged hydro-turbine applications. Furthermore, pure PLA possesses poor fatigue limits under cyclical loading, a critical failure point for continuously rotating machinery[44].

Consequently, the LLM rejects PLA and identifies PETG as a superior base polymer matrix due to its inherent chemical resilience and hydrophobicity[46]. However, pure PETG can be overly

ductile for high-load impeller applications. To address the requirement for structural rigidity under heavy hydrodynamic loads, the AI cross-references composite materials and selects Carbon Fiber reinforced PETG (PETG-CF).

The incorporation of milled or chopped carbon fibers into the PETG matrix dramatically increases the material's stiffness, dimensional stability, and resistance to creep under continuous load, while retaining the base polymer's inherent toughness, UV resistance, and water resistance[47]. The AI's output explicitly recommends PETG-CF as the optimal choice, fulfilling the mandate to provide exact material specifications.

Aerodynamic and Hydrodynamic Efficiency Constraints

A robust engineering AI must look beyond bulk material strength and address the specific physical manifestations of the chosen manufacturing process. FDM 3D printing creates parts by extruding plastic layer by layer. This inherently leaves distinct layer lines, resulting in a high degree of macroscopic surface roughness across the outer mold line of the blade.

The AI's vector search of the aerodynamic database reveals that blade surface roughness is one of the most critical factors affecting turbine aero-thermal and hydrodynamic performance[51]. Fluid dynamics research indicates that surface roughness accelerates the boundary layer transition from laminar flow to turbulent flow[52]. On a turbine blade, roughness elements disrupt the flow field, increasing parasitic friction drag and severely reducing the lift-to-drag ratio.

Experimental data retrieved by the RAG system demonstrates that at chord Reynolds numbers ($Re_c = u_c/\nu$) ranging from 0.8×10^6 to 4.8×10^6, distributed surface roughness can decrease the maximum lift-to-drag ratio by up to 40%[52]. In extreme cases, the increase in drag due to 3D printed layer lines can lead to a negative performance curve, inducing dynamic stall and rendering the turbine aerodynamically inefficient[53]. The skewness of the roughness (whether the height profile is dominated by negative valleys or positive peaks) further complicates loss generation[54].

Prescribing Post-Processing Directives

To counteract this severe hydrodynamic penalty, the AI tool does not simply stop at material selection. It recommends a mandatory post-processing step to alter the surface physics of the blade.

The system dictates that the 3D-printed PETG-CF blade must be coated with a UV-curable epoxy resin (such as the commercially available XTC-3D formulation)[55]. Applying an epoxy coating acts as a self-leveling sealant. This surface treatment fills the negative skewness valleys of the layer lines, achieving a hydraulically smooth outer surface[54]. This smoothing mitigates premature boundary layer transition, eliminates the 40% drag penalty, and provides a secondary, impenetrable barrier against deep water ingress, further protecting the internal infill structure of the print[54].

By outputting this comprehensive analysis—selecting PETG-CF, warning of abrasive nozzle wear, highlighting the structural limits, and prescribing epoxy smoothing for hydrodynamic efficiency—the system proves it can successfully emulate a multidisciplinary engineering consultant, operating entirely offline, fulfilling the capstone's ultimate objective.

Comprehensive Four-Month Execution Roadmap

To successfully deliver this exhaustive mechanical and software engineering project by the March 2027 deadline, a highly structured, agile development roadmap is required. The project spans exactly four working months (October, November, January, and February), acknowledging the December holiday hiatus where no academic progress is expected.

Month 1: October – Core Infrastructure and Knowledge Ingestion

The inaugural month is dedicated to establishing the fundamental software architecture and building the theoretical engineering database.

Week	Milestone	Technical Deliverables
Week 1-2	Environment & LLM Deployment	Configure Python virtual environments. Install LangChain, ChromaDB, and Streamlit[28]. Deploy llama.cpp to run a quantized local LLM (e.g., Llama-3 8B Q4_K_M) ensuring VRAM usage remains under 8GB for laptop compatibility[24].
Week 3-4	Knowledge Base Curation & Vectorization	Digitally scrape and format material data sheets for PLA, ABS, PETG, and CF composites. Digitize Ashby performance index methodologies[16]. Process all text through a local embedding model (sentence-transformers) and populate the persistent ChromaDB vector store[28].

Month 2: November – RAG Pipeline Integration and NLP Translation

The second month bridges the vector database with the LLM, focusing on translating layman terminology into precise engineering parameters.

Week	Milestone	Technical Deliverables

Week 1-2	LangChain Orchestration	Write backend logic connecting user inputs to ChromaDB cosine similarity search[35]. Develop the NLP translation layer prompt templates (mapping "heavy load" to yield strength parameters).
Week 3-4	Guardrails and Command Line Testing	Implement strict system prompts constraining the LLM to rely *only* on retrieved context to eliminate hallucinations[32]. Run extensive CLI queries to evaluate the accuracy of metallurgical recommendations and adjust document chunking overlap for better retrieval.

(December: Scheduled Project Hiatus)

Month 3: January – User Interface and Case Study Validation

Following the holiday break, the focus shifts to user accessibility and rigorous mechanical validation of the system's outputs.

Week	Milestone	Technical Deliverables
Week 1-2	Streamlit GUI Development	Construct the frontend interface using Streamlit and streamlit-chromadb-connection[28]. Implement the Dual-Mode toggle switch (Environment variable swap between local llama.cpp endpoint and OpenAI API endpoint). Format LLM outputs to generate Markdown tables for clear data presentation.

Week 3-4	Micro-Hydro Case Study Execution	Input complex, multi-constraint variables encompassing water saturation, UV exposure, cyclical fatigue, and FDM layer lines[5]. Force the system to evaluate PETG-CF against standard polymers and analyze its post-processing recommendations for aerodynamic surface roughness[52].

Month 4: February – Final Optimization and Thesis Generation

The final month is dedicated to polishing the software, optimizing latency, and compiling the exhaustive academic documentation required for a fifth-year capstone.

Week	Milestone	Technical Deliverables
Week 1-2	Edge-Case Testing & Optimization	Stress-test the application with obscure prompts to monitor failure modes. Optimize local inference generation speeds by adjusting CPU thread counts and GPU offloading layers within the llama.cpp configuration[28].
Week 3-4	Documentation and System Freeze	Finalize the capstone thesis. Document the software architecture, the mechanical engineering theoretical framework (Ashby methods), and the socio-economic implications for localized manufacturing in Kenya. Lock all code repositories for the March evaluation.

Conclusion

The convergence of artificial intelligence and localized additive manufacturing holds the potential to radically accelerate infrastructure development and maintenance in emerging economies. By transitioning from centralized, import-reliant supply chains to decentralized 3D printing, regions like rural Kenya can achieve true technological self-reliance in maintaining critical renewable energy systems. However, the sheer complexity of material science, combined with the catastrophic safety risks associated with standard AI hallucinations, presents a significant barrier to non-technical operators.

By constructing a dual-mode, locally hosted LLM augmented by a ChromaDB-backed Retrieval-Augmented Generation pipeline, this project delivers a secure, offline-capable expert system. This architecture successfully translates layman requirements into rigorous Ashby-style material selections, completely bypassing the need for complex manual chart analysis. When validated against the grueling mechanical and fluid dynamic parameters of a 3D-printed micro-hydro turbine blade, the system demonstrates profound utility. It successfully balances tensile strength, cyclic fatigue limits, environmental degradation, and aerodynamic surface constraints, outputting precise manufacturing directives—such as the utilization of PETG-CF paired with epoxy post-processing to eliminate friction drag. Executed over a highly disciplined four-month roadmap, this initiative not only fulfills all fifth-year academic requirements but produces a highly deployable, transformative tool that empowers localized engineering across the African continent.

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